

System of Linear Equations

I. GETTING STARTED

“Suppose you bought pens and pencils.”

- 2 pens + 1 pencil = ₹5
- 1 pen + 1 pencil = ₹3

Question is: *Can we find the price of one pen and one pencil?*

The above can be written as

$$\begin{aligned}2x + y &= 5 \\x + y &= 3\end{aligned}$$

(x = pen, y = pencil)

Step 1: Solve Without Any Formula (Basic Elimination)

We already have:

$$\begin{aligned}2x + y &= 5 \\x + y &= 3\end{aligned}$$

Subtract equations

$$\begin{aligned}(2x + y) - (x + y) &= 5 - 3 \\x &= 2\end{aligned}$$

Substitute back

$$\begin{aligned}x + y &= 3 \\2 + y &= 3 \Rightarrow y = 1\end{aligned}$$

So:

- Pen = ₹2
- Pencil = ₹1

Step 2: Now Connect to Matrices

Say:

“What if we had 100 equations? We need a structured way.”

So we write the same thing in a compact form:

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Step 3: Explain What This Means

Break it down:

- First row $\rightarrow 2x + 1y = 5$
- Second row $\rightarrow 1x + 1y = 3$

Matrix is just a **compressed way of writing equations**

Step 4: Key Idea

"We want to *undo* this matrix to get x and y."

So we write:

$$AX = B$$

We want:

$$X = ?$$

Step 5: Inverse

"Just like division reverses multiplication, inverse matrix reverses matrix multiplication."

So:

$$X = A^{-1}B$$

Step 6: Plugging in the Values

Lets say, you have got...

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Now multiply:

$$X = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Compute

$$\begin{aligned} x &= 5 - 3 = 2 \\ y &= -5 + 6 = 1 \end{aligned}$$

II. Finding Inverses

Problem I

Goal

We want to find the inverse of:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

Step 1: What does “Inverse” mean?

“Inverse of a matrix is something that gives identity when multiplied.”

$$A \cdot A^{-1} = I$$

Where:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Identity matrix = “1” of matrices

Step 2: Write Augmented Matrix (Most Intuitive Method)

We *attach identity* to A:

$$\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

Left = original matrix

Right = identity

Goal: Convert **left** → **identity**, then **right becomes inverse**

Step 3: Make First Pivot = 1

Divide Row1 by 2:

$$R1 \rightarrow \frac{R1}{2}$$
$$\begin{bmatrix} 1 & 0.5 & 0.5 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

Step 4: Make Below Element Zero

$$R2 \rightarrow R2 - R1$$
$$\begin{bmatrix} 1 & 0.5 & 0.5 & 0 \\ 0 & 0.5 & -0.5 & 1 \end{bmatrix}$$

Step 5: Make Second Pivot = 1

$$R2 \rightarrow \frac{R2}{0.5}$$
$$\begin{bmatrix} 1 & 0.5 & 0.5 & 0 \\ 0 & 1 & -1 & 2 \end{bmatrix}$$

Step 6: Make Above Element Zero

$$R1 \rightarrow R1 - 0.5 \times R2$$
$$\begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{bmatrix}$$

Final Result

Left side = Identity

Right side = Inverse

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

[Now Connect Back

$$X = A^{-1}B$$

So this inverse is what helps solve equations systematically.]

Problem 2

Find the inverse of:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

Step 1: Augment with Identity

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{bmatrix}$$

Goal: Left $\rightarrow$ Identity, Right $\rightarrow$ Inverse

Step 2: Make below pivot zero

$$R2 \rightarrow R2 - 2R1$$

$$R3 \rightarrow R3 - R1$$

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & -2 & 7 & -1 & 0 & 1 \end{bmatrix}$$

Step 3: Make below second pivot zero

$$R3 \rightarrow R3 + 2R2$$

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 9 & -5 & 2 & 1 \end{bmatrix}$$

Step 4: Make third pivot = 1

$$R3 \rightarrow \frac{R3}{9}$$

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -5/9 & 2/9 & 1/9 \end{bmatrix}$$

Step 5: Eliminate above third pivot

$$R2 \rightarrow R2 - R3$$

$$R1 \rightarrow R1 - R3$$

$$\begin{bmatrix} 1 & 2 & 0 & 14/9 & -2/9 & -1/9 \\ 0 & 1 & 0 & -13/9 & 7/9 & -1/9 \\ 0 & 0 & 1 & -5/9 & 2/9 & 1/9 \end{bmatrix}$$

Step 6: Eliminate above second pivot

$$R1 \rightarrow R1 - 2R2$$

$$\begin{bmatrix} 1 & 0 & 0 & 40/9 & -16/9 & 1/9 \\ 0 & 1 & 0 & -13/9 & 7/9 & -1/9 \\ 0 & 0 & 1 & -5/9 & 2/9 & 1/9 \end{bmatrix}$$

Final Answer

$$A^{-1} = \begin{bmatrix} 40/9 & -16/9 & 1/9 \\ -13/9 & 7/9 & -1/9 \\ -5/9 & 2/9 & 1/9 \end{bmatrix}$$

FOR EXAMPLE:

NOW THAT, We already found: A^{-1}

Now Lets take :

$$B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Key Formula

$$X = A^{-1}B$$

Step-by-Step Multiplication

$$X = \begin{bmatrix} 40/9 & -16/9 & 1/9 \\ -13/9 & 7/9 & -1/9 \\ -5/9 & 2/9 & 1/9 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

First Row (x)

$$\begin{aligned}x &= \frac{40}{9}(1) + \frac{-16}{9}(2) + \frac{1}{9}(3) \\&= \frac{40 - 32 + 3}{9} = \frac{11}{9}\end{aligned}$$

Second Row (y)

$$\begin{aligned}y &= \frac{-13}{9}(1) + \frac{7}{9}(2) + \frac{-1}{9}(3) \\&= \frac{-13 + 14 - 3}{9} = \frac{-2}{9}\end{aligned}$$

Third Row (z)

$$\begin{aligned}z &= \frac{-5}{9}(1) + \frac{2}{9}(2) + \frac{1}{9}(3) \\&= \frac{-5 + 4 + 3}{9} = \frac{2}{9}\end{aligned}$$

Final Answer

$$X = \begin{bmatrix} 11/9 \\ -2/9 \\ 2/9 \end{bmatrix}$$

JUST CONNECTING THE DOTS>....

We start with a system:

$$\begin{aligned}x + 2y + z &= 1 \\2x + 5y + 3z &= 2 \\x + 0y + 8z &= 3\end{aligned}$$

Convert to Matrix Form

$$AX = B$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Key Step

“We want to isolate X , just like solving $2x = 10 \rightarrow$ divide by 2.”

So:

$$AX = B$$

Multiply both sides by A^{-1} :

$$A^{-1}AX = A^{-1}B$$

Simplify

$$IX = A^{-1}B$$

$$X = A^{-1}B$$

Interpretation

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

This is the **entire goal of finding inverse**

WHAT YOU GUYS JUST DID

When you converted:

$$[A \mid I] \rightarrow [I \mid A^{-1}]$$

using row operations,

That process is called: **Gauss–Jordan Elimination**

What is Rank of a Matrix?

Rank = number of pivots (leading 1s) after row reduction

1. What is a Pivot?

A pivot is the first non-zero entry in a row after simplification.

Example:

$$\begin{bmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 0 \end{bmatrix}$$

Number of pivots = 2

So, **Rank = 2**

2. How to Find Rank (Procedure)

1. Take matrix A
2. Apply **Gaussian / Gauss–Jordan elimination**
3. Count number of non-zero rows (or pivots)

Method	What you do	Final form
Gaussian Elimination	Make zeros below pivots	Upper triangular matrix
Gauss–Jordan Elimination	Make zeros above & below pivots	Identity matrix

3. Why Rank Matters?

Rank tells you:

- How many **independent equations** exist
- Whether system has:
 - Unique solution
 - Infinite solutions
 - No solution

4. Key Idea

Let:

- $\text{Rank}(A)$ = rank of coefficient matrix
- $\text{Rank}([A \mid B])$ = rank of augmented matrix
- n = number of variables

THREE Fundamental Cases

Case 1: Unique Solution

$$\text{Rank}(A) = \text{Rank}([A \mid B]) = n$$

Meaning:

- Full information
- One exact solution

Example

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Rank = 2, variables = 2

Unique solution

Case 2: Infinite Solutions

$$\text{Rank}(A) = \text{Rank}([A \mid B]) < n$$

Meaning:

- Not enough independent equations
- Some variables are **free**

Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

Rank = 1, variables = 3

Infinite solutions

Case 3: No Solution

$$\text{Rank}(A) \neq \text{Rank}([A \mid B])$$

Meaning:

- Contradiction exists
- System is inconsistent

Example

$$\begin{bmatrix} 1 & 1 & | & 2 \\ 1 & 1 & | & 5 \end{bmatrix}$$

Becomes:

$$\begin{bmatrix} 0 & 0 & | & 3 \end{bmatrix}$$

No solution

Two Types of Systems

1. Consistent System

Has **solution(s)**

- One solution → **unique**
 - Many solutions → **infinite**
-

Example 1 (Unique)

$$x + y = 2$$

$$x - y = 0$$

Solution exists → **Consistent**

Example 2 (Infinite)

$$x + y + z = 3$$

$$2x + 2y + 2z = 6$$

Same equation repeated → infinite solutions → **Consistent**

2. Inconsistent System

Has **NO solution**

Example

$$x + y = 2$$

$$x + y = 5$$

Subtract:

$$0 = 3$$

Impossible → **Inconsistent**

How to Identify (Using Row Reduction)

After Gauss–Jordan / elimination:

If you get:

$$0 = 0$$

Consistent (valid equation)

If you get:

$$0 = 5$$

Inconsistent (contradiction)

Matrix View

Consistent case:

$$[0 \quad 0 \quad 0 \quad | \quad 0]$$

OK → infinite solutions

Inconsistent case:

$$[0 \quad 0 \quad 0 \quad | \quad 5]$$

Impossible → no solution

Just showing the above in terms of matrixes
(augmented form)

Case 1: Consistent System (Infinite Solutions)

Given:

$$\begin{array}{rcl} x + y + z & = & 3 \\ 2x + 2y + 2z & = & 6 \end{array}$$

Step 1: Write Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 6 \end{array} \right]$$

Step 2: Row Operation

$$R2 \rightarrow R2 - 2R1$$

Step 3: Result

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Interpretation

Last row means:

$$0x + 0y + 0z = 0 \Rightarrow 0 = 0$$

- ✓ Always true
- ✓ System is **consistent**
- ✓ Rank = 1 < variables = 3

Infinite solutions

Case 2: Inconsistent System (No Solution)

Given:

$$\begin{aligned}x + y + z &= 3 \\x + y + z &= 5\end{aligned}$$

Step 1: Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 5 \end{array} \right]$$

Step 2: Row Operation

$$R2 \rightarrow R2 - R1$$

Step 3: Result

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 2 \end{array} \right]$$

Interpretation

Last row means:

$$0x + 0y + 0z = 2 \Rightarrow 0 = 2$$

Impossible

System is **inconsistent**

No solution

LET'S TAKE AN EXAMPLE

A Simple System

$$x + y + z = 3$$

"How many variables?" $\rightarrow 3$

"How many equations?" $\rightarrow 1$

Key Insight

"One equation cannot uniquely determine three variables."

So: We expect **infinite solutions**

What is a Free Variable?

A **free variable** is a variable that we are free to choose.

In this case:

We can choose:

$$z = t \text{ (free variable)}$$

Express Other Variables

From:

$$x + y + z = 3$$

Substitute $z = t$:

$$\begin{aligned} x + y + t &= 3 \\ x &= 3 - y - t \end{aligned}$$

Final General Solution

$$(x, y, z) = (3 - y - t, y, t)$$

What This Means

For every choice of y and t , you get a valid solution

Rewriting this in a Standard Way

Let:

$$y = s, z = t$$

Then:

$$(x, y, z) = (3 - s - t, s, t)$$

“We are not finding one answer—we are describing ALL possible answers.”

Connecting this, to Rank

Lets say, we

- Variables = 3
- Rank = 1

Number of free variables:

$$3 - 1 = 2$$

So we got **2 free variables (s, t)**

General Rule

$$\text{Free variables} = n - \text{Rank}$$

More Structured Example (Using Matrix)

Given system:

$$\begin{aligned}x + y + z &= 3 \\ 2x + 2y + 2z &= 6\end{aligned}$$

Step 1: Convert to matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 6 \end{array} \right]$$

Step 2: Row reduce

$$\begin{aligned}R2 &\rightarrow R2 - 2R1 \\ \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]\end{aligned}$$

Interpretation

- Rank = 1
- Variables = 3
- Free variables = 2

Identify Free Variables

From:

$$x + y + z = 3$$

Let:

$$y = s, z = t$$

Solve:

$$x = 3 - s - t$$

Thus, we get Final Answer

$$(x, y, z) = (3 - s - t, s, t)$$

“Less rank = less information → more freedom → more solutions”
Is the above one Consistent ? Question to you all....

